

POAMS LECTURES A

Lecture 1: Introduction and Historical Background I, 7 pages.

General introduction, historical background and context I.

Lecture 2: Historical Background II, 9 pages.

Historical background and context II.

Lecture 3: Philosophical Background: Normal Realism, 9 pages.

God's-eye-view Realism, Subjective Idealism, the rise and fall of the 'Linguistic' movement, Normal Realism, light, Atomism versus Holism.

Lecture 4: The Universal Constant c , 11 pages.

Normal Realism and light, the constant c , electromagnetism and Maxwell's equations, the luminiferous ether, the Normal Realist approach to the constant c , light as pure information.

Lecture 5: Uniform Motion and Time-Dilation, 10 pages.

Uniform motion, the Normal Realist approach to uniform motion, time-dilation in relative uniform motion, experimental evidence for time-dilation, the 'cone model'.

Lecture 6: Uniform Motion: Some Consequences of Time-Dilation and the 'Cone Model', 11 pages.

Time-dilation in uniform motion, the 'cone model', the non-paradox of time-delayed instantaneity, the Doppler effect, the 'twins paradox'.

Lecture 7: Uniform Motion: Length Contraction, Minkowski Space-Time and Lorentz Transformations, 10 pages.

Time-dilation in uniform motion, length contraction, Minkowski space-time, the special Lorentz transformation, length contraction again.

Lecture 8: Uniform Motion: Relativistic Speeds and Relativistic Kinematics, 7 pages.

The special Lorentz transformation, relativistic speeds, and some relativistic kinematics.

Lecture 9: Uniform Motion: Relativistic Linear Momentum, Mass and Energy, 9 pages.

Some relativistic dynamics, classical work and energy, and relativistic mass and energy.

Lecture 10: The Ultimately Discrete Nature of Physical Phenomena, 9 pages.

The quantisation of physical measures, the origins of Quantum Mechanics, the Balmer-Rydberg formula and Bohr's model of the atom, the first stage in the development of Quantum Mechanics.

Lecture 11: Quantum Mechanics and the Discrete Nature of Light-Energy, 9 pages.

The quantisation of physical measures, the EPR controversy revisited, De Broglie waves, the second stage in the development of Quantum Mechanics, the discrete nature of light-energy.

Lecture 12: The Discrete Nature of Light-Energy from the Normal Realist Perspective, 8 pages.

The discrete nature of light-energy, causality at the quantum level, photum interaction and conservation of four-momentum, a Normal Realist derivation of the Balmer-Rydberg formula.